Assignment 11 - Project on topic Monitor and Manage Linux Processes in Linux

Creating a lab project on the topic "Monitor and Manage Linux Processes in RHEL 9" involves a hands-on approach where students can gain practical experience in working with processes in a Red Hat Enterprise Linux 9 environment.

- Red Hat Enterprise Linux 9 installed (either on a physical machine or a virtual machine).
- Access to a terminal or command line interface.

Questions for this assignment

1. Define below task with example.
   Task 1: Understanding Processes
2. Define below task with example.
   Task 2: Managing Processes
3. Define below task with example.
   Task 3: Monitoring Process Resources
4. Define below task with example.
   Task 4: Process Priority and Niceness

Solution by the Instructor:

1. Define below task with example.
   Task 1: Understanding Processes
   1.1. Open a terminal and execute the following command to display information about the currently running processes:
   ps aux
   1.2. Explain the output of the ps aux command. What information does it provide about each process?
   The ps aux command displays a detailed list of currently running processes. The columns include:
   - USER: The user who owns the process.
   - PID: The process ID.
   - %CPU: The CPU usage by the process.
   - %MEM: The memory usage by the process.
   - VSZ: The virtual memory size of the process.

- RSS: The resident set size, indicating the non-swapped physical memory used by the process.
- TTY: The terminal type associated with the process.
- STAT: The process status (e.g., R for running, S for sleeping).
- START: The start time of the process.
- TIME: The total accumulated CPU time used by the process.
- COMMAND: The command that started the process.

2. Define below task with example.

   Task 2: Managing Processes

   2.1. Identify the PID of a specific process using the ps aux | grep <process_name> command.

   ps aux | grep firefox

   This command will display information about the Firefox process, and you can identify the PID from the output.

   2.2. Use the kill command to terminate the identified process.

   kill <PID>

   Replace <PID> with the actual PID of the process obtained in Task 2.1.

3. Define below task with example.

   Task 3: Monitoring Process Resources

   3.1. Use the top command to monitor real-time information about system processes.

   top

   Explain the information displayed by the top command. What key commands can be used to interact with the top utility?

   Answer 3.1 Explanation: The top command displays real-time information about system processes. Key commands include:

   - q: Quit top.
   - k: Kill a process.
   - Spacebar: Update the display.

4. Define below task with example.

   Task 4: Process Priority and Niceness

   4.1. Use the nice command to run a process with a specific niceness value.

nice -n <niceness_value> <command>
Replace <niceness_value> with the desired niceness value and <command> with the command to be executed.
4.2. Check the niceness value of a running process using the ps command.
Answer 4.2:
ps -p <PID> -o nice
Replace <PID> with the process ID you want to check.